Classification of States

Definition. Accessibility

State j is **accessible** from state i, denoted by $i \to j$, if

$$p_{ij}^{(n)} > 0 \quad \text{for some } n \geq 0.$$

Remark.

- Accessibility is not a one-step concept: p_{ij} may be zero while $p_{ij}^{(n)} > 0$ for some $n > 1$.
- It is not symmetric: $i \to j$ does not imply $j \to i$.
- Since $p_{ii}^{(0)} = 1$, every state is accessible from itself.

Definition. Communication

States i and j **communicate**, denoted by $i \leftrightarrow j$, if

$$i \to j \qquad \text{and} \qquad j \to i.$$

Note. Communication is an **equivalence** relation.

Definition. Communication Class

A **communication class** is a maximal set $\mathscr{C} \subseteq S$ whose states communicate with one another.

Note. Communication classes form a partition of the state space S.

Definition. Irreducible chains

A Markov chain is **irreducible** if it has exactly one communication class. Otherwise, it is **reducible**.

Definition. Closed Class

A communication class $\mathscr{C}$ is **closed** if the chain cannot leave it:

$$p_{ij} = 0 \qquad \text{for every } i \in \mathscr{C} \text{ and } j \notin \mathscr{C}.$$

Remark.

- Once $X_n \in \mathscr{C}$ for any n, all future states X_{n+m} remain in $\mathscr{C}$ with probability 1, any $m \geq 0$.
- An **absorbing state** i satisfies $p_{ii} = 1$. Hence $\{i\}$ is a closed class.
- A closed class may contain several states and the chain may continue moving inside it.

Theorem

Every finite-state Markov chain has at least one closed communicating class.

Definition. Hitting and First-Return Times

For a state j, define

$$T_j = \inf\{n \geq 0 : X_n = j\}.$$

For a process **starting from state** i, define the **first *return* time**

$$T_i^+ = \inf\{n \geq 1 : X_n = i\}.$$

Remark.

- T_j: the first time to arrive at state j.
- T_i^+: we know that we start from i, and we wonder the first return time to i.

Definition. First-Return Probability

Starting from i, the probability that the first return occurs exactly at time n is denoted as

$$f_{ii}^{(n)} = P_i(T_i^+ = n), \qquad n \geq 1.$$

The probability of ever returning is

$$f_{ii} = P_i(T_i^+ < \infty) = \sum_{n=1}^{\infty} f_{ii}^{(n)}.$$

Takeaway. Consider state i. If the process returns to i almost surely (with prob. 1), i.e. $f_{ii} = 1$, then it will return to i infinitely many times, and the total returning prob. $\sum_{n=1}^{\infty} p_{ii}^{(n)}$ is infinite. We call i **recurrent**. This is a class property.

Definition. Recurrence and Transience

A state i is **recurrent** if

$$P_i(T_i^+ < \infty) = 1.$$

It is **transient** if

$$P_i(T_i^+ < \infty) < 1.$$

Define the number of visits to i after time zero:

$$N_i = \sum_{n=1}^{\infty} \mathbf{1}\{X_n = i\}.$$

Theorem. Number of Revisits

For every $m \geq 1$,

$$P_i(N_i \geq m) = (f_{ii})^m.$$

Consequently,

$$\begin{cases} N_i = \infty \text{ a.s.}, & f_{ii} = 1, \\ N_i < \infty \text{ a.s.}, & f_{ii} < 1. \end{cases}$$

$$E_i[N_i] = \begin{cases} \dfrac{f_{ii}}{1 - f_{ii}}, & f_{ii} < 1, \\ \infty, & f_{ii} = 1. \end{cases}$$

Hence,

$$i \text{ recurrent} \iff N_i = \infty \text{ a.s.} \iff E_i[N_i] = \infty,$$

$$i \text{ transient} \iff N_i < \infty \text{ a.s.} \iff E_i[N_i] = \frac{f_{ii}}{1 - f_{ii}}.$$

Define the n-step return probability

$$p_{ii}^{(n)} = P_i(X_n = i).$$

Theorem. Return-Probability Criterion

$$i \text{ is recurrent} \iff \sum_{n=1}^{\infty} p_{ii}^{(n)} = \infty,$$

$$i \text{ is transient} \iff \sum_{n=1}^{\infty} p_{ii}^{(n)} < \infty.$$

Starting from state i,

$$\begin{aligned} i \text{ recurrent} &\iff P_i(T_i^+ < \infty) = f_{ii} = 1 \\ &\iff P_i(N_i = \infty) = 1 \\ &\iff E_i[N_i] = \infty \\ &\iff \sum_{n=1}^{\infty} p_{ii}^{(n)} = \infty. \end{aligned}$$

Remark.

- $f_{ii}^{(n)} = P_i(T_i^+ = n)$ counts paths whose *first* return occurs at time n.
- $p_{ii}^{(n)} = P_i(X_n = i)$ counts all paths that are at i at time n, including paths that returned earlier.

Note. So, $f_{ii}^{(n)} \leq p_{ii}^{(n)}$.

Theorem

If $i \leftrightarrow j$, then i and j are either both recurrent or both transient.

Theorem

For a **finite-state** Markov chain:

- at least **one closed communicating class** exists;
- recurrent class $\iff$ closed class;
- all states in a nonclosed class are transient.

Definition. Period

For a state i, define

$$d(i) = \gcd\{n \geq 1 : p_{ii}^{(n)} > 0\}.$$

If the set is empty, use the convention $d(i) = 0$.
A state is **aperiodic** if $d(i) = 1$.

Theorem

If $i \leftrightarrow j$, then

$$d(i) = d(j).$$

Definition. Aperiodic Chain

A Markov chain is **aperiodic** if every state is aperiodic, i.e. $d(i) = 1$ for all $i \in S$.

Theorem

Let state i have period $d = d(i) \geq 1$.

- For every $n \geq 1$,

$$d \nmid n \quad \Longrightarrow \quad p_{ii}^{(n)} = 0.$$

- There exists an integer $N_i \geq 1$ such that

$$p_{ii}^{(kd)} > 0 \qquad \text{for every integer } k \geq N_i.$$

Note. Some small multiples of d may not be possible return times.

Definition. Regularity

A **finite-state** Markov chain with transition matrix $\mathbf{P}$ is **regular** if there exists an integer $r \geq 1$ such that **every entry of $\mathbf{P}^r$ is strictly positive**; that is,

$$p_{ij}^{(r)} > 0 \qquad \text{for all } i, j \in S.$$

Theorem

For a finite-state Markov chain,

regular $\iff$ irreducible and aperiodic.

Theorem

Let $\{X_n\}$ be a **regular** Markov chain on a finite state space S, with transition matrix $\mathbf{P}$. There exists a distribution $\boldsymbol{\pi} = (\pi(j))_{j\in S}$ such that, for every initial distribution $\boldsymbol{\pi}_0$,

$$\lim_{n\to\infty} \boldsymbol{\pi}_n = \lim_{n\to\infty} \boldsymbol{\pi}_0 \mathbf{P}^n = \boldsymbol{\pi}.$$

- The convergence is entrywise convergence, i.e. $\pi_n(j) \to \pi(j)$ for all $j \in S$.
- This limit is independent of $\boldsymbol{\pi}_0$, i.e. $p_{ij}^{(n)} \to \pi(j)$ for all $i, j \in S$.
- $\boldsymbol{\pi}$ is unique, and it satisfies

$$\boldsymbol{\pi}\mathbf{P} = \boldsymbol{\pi}, \qquad \sum_{j\in S} \pi(j) = 1, \qquad \pi(j) > 0.$$

To calculate the limiting distribution, solve

$$\boldsymbol{\pi}\mathbf{P} = \boldsymbol{\pi}, \qquad \sum_{j\in S} \pi(j) = 1, \qquad \pi(j) \geq 0.$$

- Before calculation, **verify the convergence first!**
- This is a **linear system** in the unknowns $(\pi(j))_{j\in S}$.

Remark. The jth entry $\pi(j)$ of $\boldsymbol{\pi}$ can be interpreted as the **long-run fraction of time spent in state** j:

$$\left(\frac{1}{n}\sum_{k=1}^{n} \mathbf{1}_{\{X_k = j\}}\right)_{j\in S} \xrightarrow{\text{a.s.}} \boldsymbol{\pi}.$$

This convergence holds with probability 1.

Definition. Positive and Null Recurrence

For a recurrent state j, let $m_j = \mathbb{E}_j[T_j^+]$, where T_j^+ is the first return time to state j.

- **Positive** recurrent: $m_j < \infty$.
- **Null** recurrent: $m_j = \infty$.

Remark.

- Recurrence guarantees a return, but the mean may be infinite. Positive recurrence further guarantees a finite return time.
- Positive and null recurrence are class properties.
- Every finite irreducible chain is positive recurrent.
- Null recurrence only applies for infinite state MCs. For finite state MCs, any recurrent state is positive recurrent.

Definition. Ergodicity

A Markov chain is **ergodic** if it is irreducible, **positive recurrent**, and aperiodic.

Theorem. Limiting Distribution

An ergodic chain on a finite or countable state space S has a unique limiting distribution $\boldsymbol{\pi} = (\pi(j))_{j\in S}$. For every initial distribution $\boldsymbol{\pi}_0$,

$$\lim_{n\to\infty} \pi_n(j) = \pi(j) = \frac{1}{m_j} > 0, \qquad j \in S.$$

- The limit is independent of $\boldsymbol{\pi}_0$.
- It satisfies

$$\boldsymbol{\pi}\mathbf{P} = \boldsymbol{\pi}, \qquad \sum_{j\in S} \pi(j) = 1.$$

- For finite chains, ergodicity is equivalent to regularity.
- Irreducible + aperiodic + finite states ⇒ limiting dist.
- Irreducible + aperiodic + positive recurrent ⇒ limiting dist.
- **Convergence.** Finite chains: regular, equivalently irreducible and aperiodic. Countable chains: irreducible, positive recurrent and aperiodic. Then $\boldsymbol{\pi}_n \to \boldsymbol{\pi}$, independently of $\boldsymbol{\pi}_0$.
- **Calculation.** Verify convergence first; then solve $\boldsymbol{\pi}\mathbf{P} = \boldsymbol{\pi}$, $\sum_j \pi(j) = 1$, $\pi(j) \geq 0$. For an ergodic chain, $\pi(j) = 1/m_j$.
- **Interpretation.** In an ergodic chain, fixed-time probabilities and one-path visit frequencies both approach $\pi(j)$.
- **Periodicity and null recurrence.** Positive recurrent periodic chains retain phase limits and time averages, including one-path frequencies. For a null recurrent state j, $\pi_n(j) \to 0$.
- **Multiple classes.** Combine entering probabilities with within-class behaviour. The starting state can affect the limit.

	Transient	Null recurrent	Positive recurrent
$\mathbb{P}_i(T_i^+ < \infty)$	$f_{ii} < 1$	1	1
N_i	finite a.s.	∞ a.s.	∞ a.s.
$m_i = \mathbb{E}_i[T_i^+]$	∞	∞	finite
$\mathbb{E}_i[N_i]$	$f_{ii}/(1-f_{ii})$	∞	∞
Long-run fraction	0	0	$1/m_i > 0$

Shared Properties Within a Class

Property	Shared throughout a communicating class?
Recurrent or transient	Yes
Positive recurrent or null recurrent	Yes
Period	Yes
Stationary probability value $\pi(i)$	Not necessarily
Expected return time $\mathbb{E}_i[T_i^+]$	Not necessarily
Individual transition probabilities	Not necessarily
Exact hitting times	Not necessarily
Closedness	A property of the whole class

Stationary Distributions

Definition. Stationary Distribution

A probability vector $\boldsymbol{\pi} = (\pi(j))_{j\in S}$ is a **stationary distribution** for $\mathbf{P}$ if $\boldsymbol{\pi} = \boldsymbol{\pi}\mathbf{P}$.

Definition. Global Balance Equations

A probability vector $\boldsymbol{\pi}$ satisfies **global balance** if, for every j,

$$\underbrace{\pi(j)}_{\text{Leaving prob.}} = \underbrace{\sum_i \pi(i) p_{ij}}_{\text{Arriving prob.}}.$$

Equivalently, excluding self-transitions, $\pi(j)\sum_{k\neq j} p_{jk} = \sum_{i\neq j} \pi(i) p_{ij}$.

Definition. Detailed Balance Equations

A probability vector $\boldsymbol{\pi}$ satisfies the **detailed balance equations** if, for every pair i, j,

$$\pi(i) p_{ij} = \pi(j) p_{ji}.$$

Note. Detailed balance ⇒ global balance, but the converse is not true.

Theorem

For an **irreducible** chain, the existence of a stationary probability distribution implies **positive recurrence**.

Condition for Unique Stationary Distribution

Chain structure	Is the stationary distribution π_* unique?	Does $\pi_n = \pi_0 P^n$ converge?	Is the limit independent of π_0?
Finite, irreducible, aperiodic	Yes	Yes	Yes
Finite, irreducible, periodic	Yes	Generally no	No single full limit
Finite, reducible, exactly one closed class, aperiodic	Yes	Yes	Yes
Finite, reducible, exactly one closed class, periodic	Yes	Generally no	No single full limit
Finite, reducible, multiple closed classes, all aperiodic	No	Yes	Generally no
Finite, reducible, multiple closed classes, some periodic	No	May fail	Generally no
Countable, irreducible, positive recurrent, aperiodic	Yes	Yes	Yes
Countable, irreducible, positive recurrent, periodic	Yes	Generally no	No single full limit
Countable, irreducible, null recurrent	No stationary probability distribution	No	Not applicable
Countable, irreducible, transient	No stationary probability distribution	No	Not applicable

For an ergodic chain—irreducible, positive recurrent, and aperiodic—the limit is independent of the initial distribution π_0 and equals the unique stationary distribution π_*. A stationary distribution can still be unique when the chain is periodic; periodicity affects convergence, not uniqueness.

Markov Chain Monte Carlo

Definition. Metropolis–Hastings Construction

Given the target $\boldsymbol{\pi}$ on S, construct $\mathbf{P}$ in three steps.

1. **Choose a proposal matrix Q** on S: transition probability matrix, irreducible
2. **Specify the acceptance probabilities:**

$$\alpha(i,j) = \min\left\{1, \frac{\pi_j q_{ji}}{\pi_i q_{ij}}\right\} \quad (i \neq j,\ q_{ij} > 0).$$

3. **Construct the transition prob. matrix P:**

$$p_{ij} = q_{ij}\alpha(i,j) \quad (j \neq i), \qquad p_{ii} = 1 - \sum_{j \neq i} p_{ij}.$$

This specifies the law of the MC

Definition. Metropolis–Hastings Implementation

Choose $X_0 \in S$. Given $X_n = i$, generate one step as follows.

1. Draw a proposal Y_n with $\mathbb{P}(Y_n = j \mid X_n = i) = q_{ij}$.
2. Independently draw $U_n \sim \text{Uniform}(0,1)$.
3. Set

$$X_{n+1} = \begin{cases} Y_n, & U_n < \alpha(i, Y_n), \\ i, & U_n \geq \alpha(i, Y_n). \end{cases}$$

4. Record X_{n+1}, and continue.

Metropolis-Hastings algorithm:

- The acceptance probability follows that

$$\alpha(i,j) = \min\left\{1, \frac{w_j q_{ji}}{w_i q_{ij}}\right\}.$$

  No calculation of C.
- The target must still be normalizable: $Z < \infty$ is required.

Now $\boldsymbol{\pi}$ satisfies the detailed balance equations, so $\boldsymbol{\pi}\mathbf{P} = \boldsymbol{\pi}$.

- **Irreducibility:** One-way proposals are rejected. Check if two-way proposals are sufficient for irreducibility.
- **Positive recurrence:** follows from irreducibility and the stationary probability distribution $\boldsymbol{\pi}$ (proof not in class)
- **Aperiodicity:** Rejection makes $p_{ii} > 0$ for one state i.

Theorem. Guarantee of Convergence

Under these conditions, for every $i, j \in S$,

$$\mathbb{P}(X_n = j \mid X_0 = i) \longrightarrow \pi_j.$$